Soil Type Classification Using Handcrafted Visual Features and Ensemble Learning

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1 Introduction

This project focuses on supervised soil-image classification using handcrafted features and classical machine learning algorithms. Instead of relying on GPU-heavy deep learning approaches, this methodology extracts statistical, texture-level, and structural descriptors from each image and evaluates multiple ML models using cross-validated Macro-F1 scores. The final classifier is constructed as a soft-voting ensemble of the best-performing models.

2 Objective

The objective is to develop an accurate, lightweight soil classification system using handcrafted visual features combined with ensemble learning.

3 Dataset Description

The dataset comprises seven distinct soil classes commonly found in various geographical regions. Each class represents a unique soil type with characteristic visual properties:

1. **Alluvial Soil:** Deposited by rivers, rich in nutrients
2. **Arid Soil:** Found in dry regions with low organic content
3. **Black Soil:** Rich in clay content, retains moisture well
4. **Laterite Soil:** Rich in iron and aluminum, reddish appearance
5. **Mountain Soil:** Found in hilly terrains with varied composition
6. **Red Soil:** Rich in iron oxide, gives characteristic red color
7. **Yellow Soil:** Contains hydrated iron oxide, yellowish appearance

3.1 Dataset Structure

The dataset is organized into the following components:

- `train/` – Directory containing labeled training images
- `test/` – Directory containing unlabeled testing images
- `train.csv` – CSV file mapping image filenames to soil type labels

3.2 Preprocessing

All images in the dataset were resized to a uniform dimension of 128×128 pixels to ensure consistency in feature extraction and to reduce computational complexity. This standardization is crucial for:

- Ensuring uniform feature vector dimensions
- Reducing memory requirements

- Accelerating feature extraction computations
- Maintaining aspect ratio consistency across the dataset

4 Feature Engineering

Feature engineering is the cornerstone of this classification approach. A comprehensive set of handcrafted features was extracted from each image, capturing various aspects of visual information. The features are categorized into color, texture, and structural groups.

4.1 Color Features

Color features capture the distribution and characteristics of color information in different color spaces:

4.1.1 RGB Color Space

- Mean values of Red, Green, and Blue channels
- Standard deviation of Red, Green, and Blue channels

4.1.2 HSV Color Space

- Mean values of Hue, Saturation, and Value channels
- Standard deviation of Hue, Saturation, and Value channels

4.1.3 LAB Color Space

- Mean values of L*, a*, and b* channels
- Standard deviation of L*, a*, and b* channels

The use of multiple color spaces ensures that various aspects of color perception are captured, as each color space emphasizes different visual properties.

4.2 Texture Features

Texture features describe the spatial arrangement of pixel intensities and are crucial for distinguishing between soil types with similar colors but different structural properties.

4.2.1 Gray Level Co-occurrence Matrix (GLCM)

GLCM features capture the spatial relationships between pixels at specific offsets:

- **Contrast:** Measures local intensity variations
- **Homogeneity:** Measures closeness of distribution to diagonal
- **Energy:** Measures uniformity of texture
- **Correlation:** Measures linear dependency of gray levels

4.2.2 Local Binary Patterns (LBP)

LBP features encode local texture patterns by comparing each pixel with its neighbors:

- LBP histograms with radius 1
- LBP histograms with radius 2
- LBP histograms with radius 3

Multiple radii ensure that both fine-grained and coarse texture patterns are captured.

4.2.3 Gabor Filters

Gabor filter responses capture oriented texture patterns:

- Mean and standard deviation of filter responses
- Four orientations: 0, 45, 90, 135

4.3 Structural and Statistical Features

These features capture global properties of the image:

- **Shannon Entropy:** Measures randomness and information content
- **Edge Density:** Computed using Canny edge detection, measures structural complexity
- **16-bin Grayscale Histogram:** Captures intensity distribution

4.4 Feature Visualization

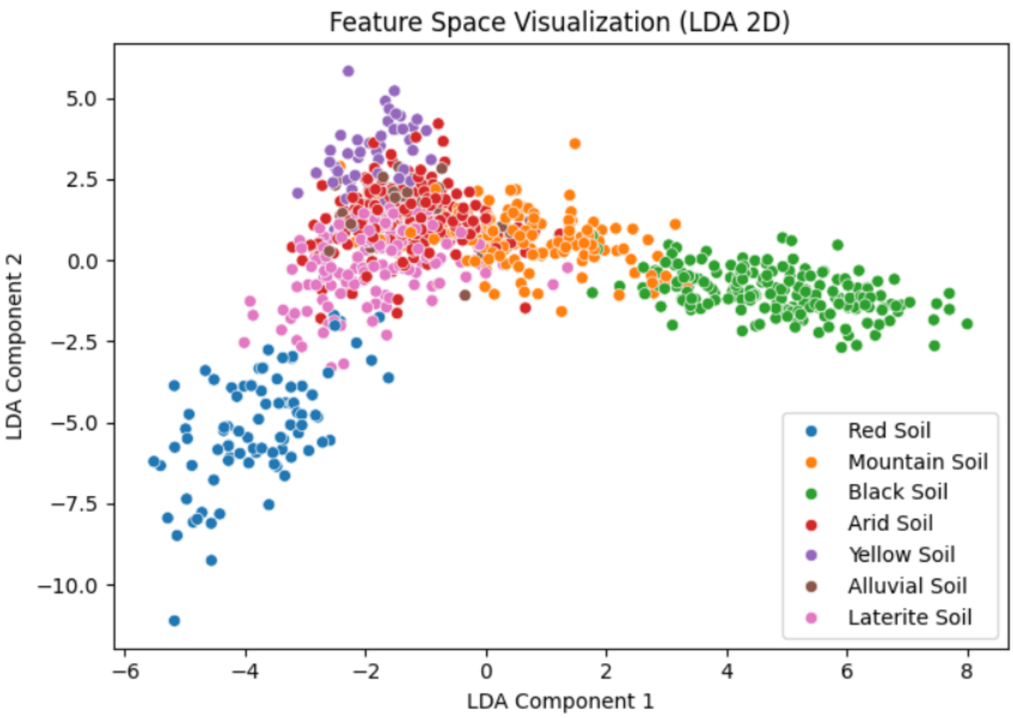

Figure 1: LDA-based 2D projection of handcrafted features showing class separability

Figure 1 shows the Linear Discriminant Analysis (LDA) projection of the extracted features into a 2D space. The visualization demonstrates that the handcrafted features provide good class separability, with distinct clusters forming for different soil types.

5 Models Implemented

Seven classical machine learning algorithms were implemented and evaluated for the soil classification task. Each model brings unique strengths to the classification problem:

5.1 Support Vector Machine (SVM)

SVM with RBF kernel creates non-linear decision boundaries by mapping features to a higher-dimensional space. It is particularly effective for complex classification problems with well-separated classes.

5.2 Random Forest Classifier

An ensemble of decision trees that reduces overfitting through bootstrap aggregating and random feature selection. Provides robust performance across various domains.

5.3 Gradient Boosting Classifier

Sequentially builds an ensemble of weak learners, with each subsequent model focusing on correcting errors of previous models. Known for high accuracy on structured data.

5.4 XGBoost Classifier

An optimized implementation of gradient boosting with regularization techniques. Offers faster training and better performance through advanced optimization algorithms.

5.5 CatBoost Classifier

Designed to handle categorical features efficiently and employs ordered boosting to reduce overfitting. Provides excellent performance with minimal hyperparameter tuning.

5.6 Logistic Regression

A linear model extended to multinomial classification. Serves as a strong baseline and provides interpretable results.

5.7 Gaussian Naive Bayes

A probabilistic classifier based on Bayes' theorem with independence assumptions. Computationally efficient and works well when feature independence holds approximately.

6 Hyperparameter Optimization

Hyperparameter optimization was performed using RandomizedSearchCV with 5-fold stratified cross-validation. Stratified folding ensures that class distributions are maintained across all folds, which is crucial for imbalanced datasets.

6.1 Optimization Strategy

The optimization process involved:

1. Defining search spaces for each model's hyperparameters
2. Performing random sampling from the search space
3. Evaluating each configuration using stratified 5-fold cross-validation
4. Selecting the configuration with the highest Macro-F1 score

6.2 Optimized Hyperparameters

The following table summarizes the best hyperparameters found for each model:

Table 1: Optimized hyperparameters for each model

Model	**Best Hyperparameters**
Random Forest	n_estimators = 400, max_depth = 20
SVM (RBF)	C = 5, gamma = 0.01, class_weight = balanced
Gradient Boosting	n_estimators = 200, learning_rate = 0.1, max_depth = 2
XGBoost	n_estimators = 200, max_depth = 4, learning_rate = 0.1
CatBoost	iterations = 400, depth = 6, learning_rate = 0.1

6.3 Key Observations

- **Random Forest:** Benefits from a large number of estimators with moderate depth
- **SVM:** Balanced class weights help handle any class imbalance
- **Gradient Boosting:** Shallow trees with moderate learning rate prevent overfitting
- **XGBoost:** Similar configuration to Gradient Boosting with slightly deeper trees
- **CatBoost:** Larger iteration count compensates for moderate learning rate

7 Evaluation and Results

All models were evaluated using stratified 5-fold cross-validation with Macro-F1 score as the primary evaluation metric. The Macro-F1 score provides a balanced assessment of model performance across all classes, making it ideal for multi-class classification problems.

7.1 Cross-Validation Results

Table 2: Cross-validation Macro-F1 scores for all models

Model	Macro-F1 Score
Random Forest	0.7448
SVM (RBF)	**0.7882**
Gradient Boosting	0.7689
Logistic Regression	0.7802
Gaussian Naive Bayes	0.5543
XGBoost	0.7668
CatBoost	0.7746

7.2 Performance Analysis

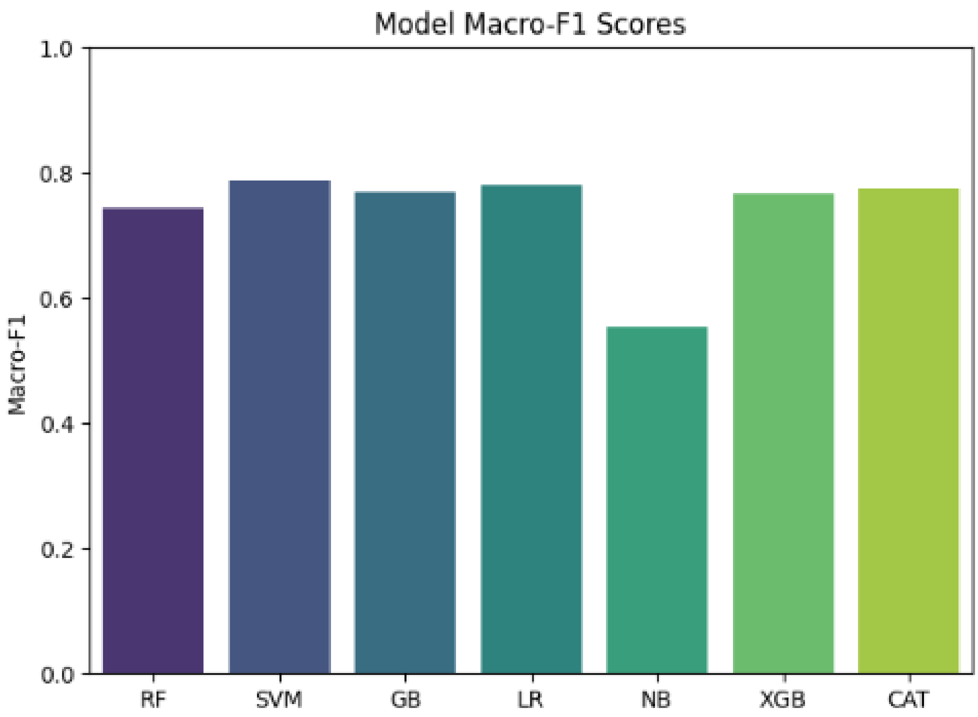

Figure 2: Comparison of models based on Macro-F1 scores

Key findings from the evaluation:

- **SVM (RBF)** achieved the highest individual performance with a Macro-F1 score of 0.7882
- **Logistic Regression** performed surprisingly well (0.7802), demonstrating that linear relationships exist in the feature space

- **Gradient Boosting** and **XGBoost** showed strong performance (0.7689 and 0.7668 respectively)
- **CatBoost** achieved competitive results (0.7746) with minimal tuning
- **Random Forest** delivered solid performance (0.7448)
- **Gaussian Naive Bayes** underperformed (0.5543), likely due to violated independence assumptions

8 Ensemble Strategy

Based on the cross-validation scores, an ensemble approach was adopted to combine the strengths of multiple models. The ensemble uses soft voting, where each model's predicted class probabilities are averaged to make the final prediction.

8.1 Model Selection

The top three models selected for the soft-voting ensemble were:

1. **SVM (RBF)** – Highest individual performance
2. **Gradient Boosting** – Strong sequential learning
3. **XGBoost** – Optimized gradient boosting with regularization

8.2 Ensemble Performance

The soft-voting ensemble achieved a training Macro-F1 score of:

0.9764

This represents a significant improvement over individual models, demonstrating the power of ensemble learning in combining diverse model predictions.

8.3 Confusion Matrix Analysis

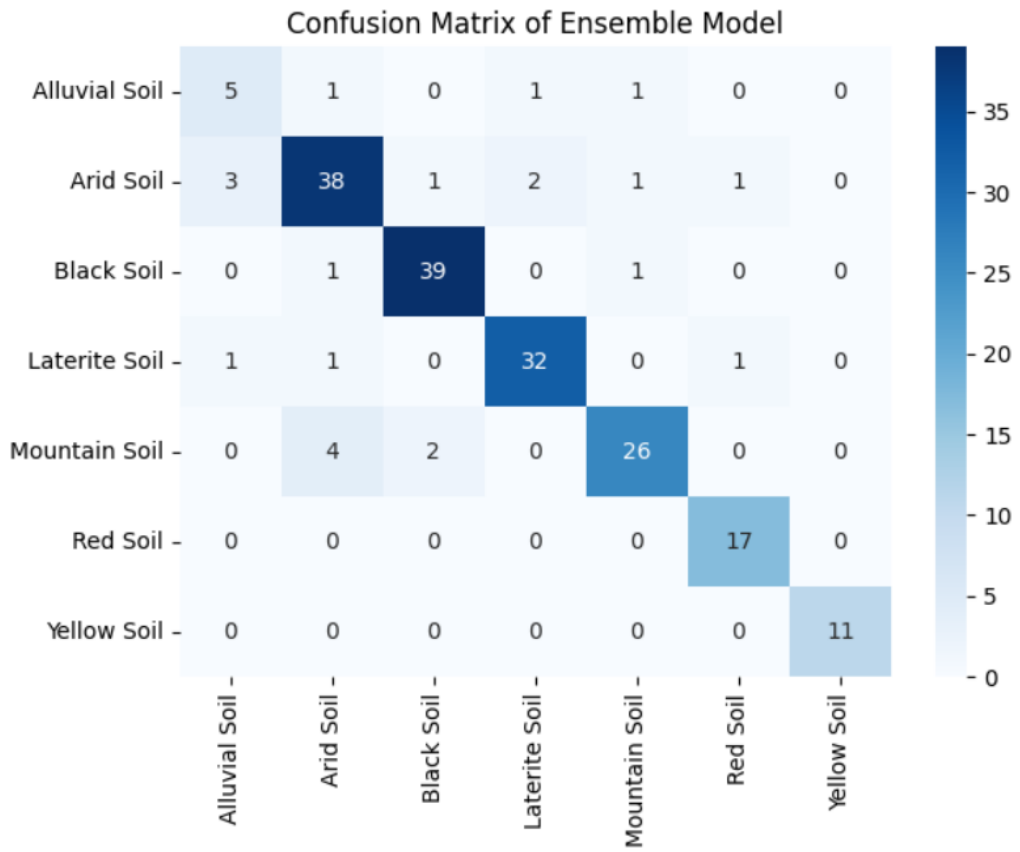

Figure 3: Confusion matrix showing classification performance across all soil types

The confusion matrix (Figure 3) reveals:

- Strong diagonal values indicating high accuracy for most classes
- Minimal confusion between distinct soil types
- Potential challenges in distinguishing between visually similar classes
- Overall balanced performance across all seven soil classes

9 Pipeline Workflow

The complete classification pipeline follows a systematic workflow:

1. **Image Preprocessing**
   - Load images from directory
   - Resize all images to 128×128 pixels
   - Normalize pixel values
2. **Feature Extraction**

- Extract color features (RGB, HSV, LAB)
- Compute texture features (GLCM, LBP, Gabor)
- Calculate structural features (entropy, edge density, histograms)

3. **Feature Standardization**
   - Convert all features to float64 format
   - Apply StandardScaler to normalize feature ranges
   - Ensure zero mean and unit variance
4. **Hyperparameter Tuning**
   - Define hyperparameter search spaces
   - Execute RandomizedSearchCV
   - Select best configurations based on cross-validation
5. **Model Evaluation**
   - Perform stratified 5-fold cross-validation
   - Calculate Macro-F1 scores
   - Compare model performances
6. **Ensemble Construction**
   - Select top-performing models
   - Configure soft-voting ensemble
   - Train ensemble on full training set
7. **Prediction and Submission**
   - Generate predictions on test set
   - Create submission CSV file
   - Validate output format

10 Sample Predictions

Table 3 shows example predictions made by the ensemble model on test images:

Table 3: Example outputs from the ensemble model

Image ID	Predicted Class
0001.jpg	Red Soil
0002.jpg	Black Soil
0003.jpg	Yellow Soil
0004.jpg	Mountain Soil
0005.jpg	Yellow Soil

11 Conclusion

This project successfully demonstrated that handcrafted visual features combined with classical machine learning methods can achieve strong performance in soil type classification. The key findings include:

- Comprehensive feature engineering is crucial for capturing visual properties of soil images
- Classical ML algorithms can effectively classify soil types without requiring GPU resources
- Ensemble methods significantly improve performance over individual models
- The soft-voting ensemble (SVM + Gradient Boosting + XGBoost) achieved a training Macro-F1 score of 0.9764